

# Rahul Saini

Amroha, Uttar Pradesh

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## INTERNSHIP

### AI/ML Developer Intern

Oct 2025 – Present

*Bitlance Tech Hub Private Limited*

- Architected and deployed multi-agent LLM pipelines using LangChain and n8n, integrating ML model inference endpoints to automate business workflows and **reduce manual operations by 40%**.
- Built real-time AI decision systems connecting ML models to production workflows, **improving automation efficiency by 30%**; designed evaluation metrics to monitor model output quality continuously.
- Established ML testing protocols covering edge-case validation, response quality benchmarking, and automated regression testing to maintain **95%+ production reliability** across all deployed models.

## PROJECTS

### Material Demand Prediction System | Python, K-Means, ARIMA

Sep 2025 – Dec 2025

- Proposed a hybrid forecasting framework combining K-Means clustering with cluster-specific time-series models to improve material demand prediction for power transmission and substation projects.
- Reduced data heterogeneity by clustering projects based on material consumption and project attributes, enabling more accurate temporal forecasting.
- Achieved significant performance gains over baseline models with **up to 35% lower MAE** and **48% improved MAPE**, supporting efficient supply chain planning and cost optimization.
- GitHub Link: <https://github.com/rahul7697762/hybrid-demand-forecast.git>

### Crop Guidance System | Python, Machine Decision Tree, Random Forest, Weather API

Aug 2025 – Oct 2025

- Designed a supervised learning crop recommendation engine using Decision Tree and Random Forest models trained on soil nutrients (N, P, K), pH, rainfall, temperature, and humidity features.
- Performed data preprocessing including missing value handling, feature scaling, categorical encoding, and feature importance analysis to improve model generalization and interpretability.
- Evaluated models using cross-validation and accuracy metrics, selecting Random Forest for deployment due to superior performance (**89% accuracy**).
- GitHub: <https://github.com/rahul7697762/Crop-Guidance-System.git>

## CERTIFICATIONS & ACHIEVEMENTS

### Certifications

- Data Structures & Algorithms (72 hrs) - Iamneo
- Object-Oriented Programming (72 hrs) - Iamneo

Dec 2025

Dec 2025

### Achievements

- Solved **600+ coding problems** across platforms including **300+ DSA problems on LeetCode** (Contest Rating: **1664**).
- Participated in **Smart India Hackathon (SIH)**, contributing to solution design and rapid prototyping.
- Ranked **11th** in a national-level web development competition by CipherSchools.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, SQL

**Machine Learning:** Data Preprocessing, Feature Engineering, Supervised Learning (Linear & Logistic Regression, Decision Trees, Random Forest, SVM, KNN, Naive Bayes), Unsupervised Learning (K-Means, Hierarchical Clustering, DBSCAN, PCA), Ensemble Methods, Model Training & Evaluation

**Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), LangChain, Intelligent Agents, Large Language Models

**Frameworks & Libraries:** Scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face, Computer Vision (CV)

**Backend & Deployment:** FastAPI, Flask, RESTful APIs, Model Inference Endpoints, Docker

**Databases & Tools:** MySQL, Vector Database, Git, GitHub, CI/CD, AWS

## EDUCATION

### Lovely Professional University

2023 – 2027

*Bachelor of Technology in Computer Science and Engineering • CGPA: 7.7*

*Phagwara, Punjab*

### A.W Faez E-Aam Inter College

2021 – 2022

*12th with Science • Percentage: 69.8%*

*Amroha, Uttar Pradesh*

### H.P Inter College

2019 – 2020

*10th with Science • Percentage: 69.7%*

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